TrainYourSnakeAI: A Novel Tool to Teach Reinforcement Learning to Middle School Students

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Abstract

Artificial Intelligence (AI) is growing rapidly in our society and is now apparent in our day-to-day lives. With the recent burst of interest in AI, many individuals and children may view AI as something mystic and magical. It is important to demystify and introduce to them how AI is made and works. To address this need, we developed a software application that allows children to specify the parameters used by a Reinforcement Learning (RL) algorithm. Then students experience how RL is used to train an AI model to play the game "Snake." This software tool was tested with 71 middle school-age students. Here, we describe the design of the *TrainYourSnakeAI* application, the approach we used to introduce the associated ideas to middle school children, and how we assessed student learning. Qualitative data collected from students are presented and discussed. We surveyed their knowledge of AI before and after using the application. In this work, our research questions were: (RQ1) How can we create an engaging tool to teach reinforcement learning? and (RQ2) Does using our application foster a stronger understanding of reinforcement learning in children? Our findings indicate that students were able to understand the functionality of reward functions and how agents can learn from the environment using the concept of RL. We found that out of the 51 students who were not previously familiar with RL, 40 were able to provide adequate descriptions of RL after using *TrainYourSnakeAI*.

CCS Concepts

• **Social and professional topics** → **K-12 education**; • **Human-centered computing** → **Interactive systems and tools**; • **Applied computing** → **Interactive learning environments**; • **Computing methodologies** → *Knowledge representation and reasoning*.

*Cesar and Priyanka contributed equally to this research.

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Keywords

Reinforcement learning, reward function, AI literacy, game-based learning, qualitative and quantitative data analysis

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1 Introduction and Motivation

In today's rapidly evolving world, artificial intelligence (AI) is becoming an integral part of daily life [10, 19], influencing even our children's early interactions through platforms like social media, games, Alexa, Bixby, and Siri, etc. Thus, the younger generation needs to understand, assess, and familiarize themselves with AI systems [11, 24]. This knowledge can help them evaluate information, improve computational thinking skills, and prepare for future job markets [2, 12].

AI education can foster critical thinking and problem-solving skills. With the usage of AI concepts, students can learn to analyze complex problems, break them down into manageable components, and devise creative solutions. This analytical mindset nurtures their ability to adapt to various challenges, which is a crucial skill in an era defined by uncertainty and rapid change. In addition, AI education cultivates digital literacy and technological fluency. In an increasingly digital world, understanding the fundamentals of AI empowers students to navigate and harness technology effectively. By exploring AI applications and concepts, students are inspired to think outside the box and develop innovative solutions to real-world problems.

In this paper, we describe the design and implementation of the *TrainYourSnakeAI* game for Artificial Intelligence (AI) education and more specifically on Reinforcement Learning (RL). Our application allows middle school students to learn about RL by training an agent to play the game Snake. Students can see how their training

data causes an agent to learn from the environment. The idea behind this is that through our web-based application, every student gets the opportunity to become a future-ready AI-literate citizen.

2 Related Work

Integrating AI education into K-12 education is becoming increasingly necessary due to how AI is becoming more commonly used in various fields. For example, AI in the education field can help personalize learning, letting students learn at their speed and meeting their specific needs.

As such, AI researchers for K-12 working group developed an online resource directory for teachers with access to AI-related videos, demonstrations, software, and activity descriptions that they can integrate into their lesson plans [13, 22] and designed AI technologies centered on learners that foster increased user understanding of AI [12]. Recently, Shuchi Grover introduced the early AI and CS education efforts in K-12 AI education, highlighted critical design issues, and offered guidance for future initiatives to effectively integrate AI learning into K-12 curricula [3, 8].

In the literature, many game-based applications [13, 29] and tools [14] were introduced, like a social robot-based application [5], game-based approach [18, 25], and a block-based programming tool [7] to teach students aged 10–16 with focused AI concepts through interactive and project-based learning activities [15] in K-12 education.

There has been a growing interest in teaching Reinforcement Learning (RL) specifically at various educational levels, from elementary school to undergraduate studies. The AI4K12 guidelines emphasize the importance of "Learning" within the "Five Big Ideas in AI," including reinforcement learning as a key machine learning concept for grades 9-12, where students learn how algorithms adjust internal models through environmental interaction and feedback, highlighting the necessity of teaching RL to K-12 students [1]. Many researchers have explored methods to introduce RL using accessible tools and activities tailored for teenagers [15]. Furthermore, RL can also be used in various applications and fields which demonstrates the increasing significance of AI literacy [23, 27].

In recent years, RL has been introduced through digital game designs. For example, Voulgari et al.'s work on *ArtBot* teaches students at the primary and middle school level to explore the impact of parameters, rewards, and training data on the learned behavior of AI agents through these games [25, 28]. The *ArtBot* game is designed to introduce two important concepts: supervised learning and RL. The first part of this game introduces supervised learning by allowing the player to train an AI helper to find art objects. During the second part of the game, players, along with their AI helper, navigate dungeons to collect stolen art objects, learning RL concepts by assigning rewards to objects and avoiding traps, adjusting parameters like exploration and exploitation rates, and observing the AI's behavior using a Q-learning algorithm on an 8x8 grid while modifying reward settings with sliders. This work demonstrates an interactive approach to introduce children to RL.

Quiloan et al.'s work on the *PlayGrid* project developed three prototypes to teach RL to children, with a focus on creating tools that can benefit young children who may not have coding knowledge [18]. *PlayGrid*'s final prototype involved a hardware device where children could view an RL agent navigating through a grid while avoiding obstacles and eating food. This work demonstrates that visualization of RL can be a good way to introduce children to AI.

Dietz et al. created *ARtonomous*, an iPad application that introduced middle school students to RL using virtual robotics [5]. *ARtonomous* allows children to view virtual autonomous vehicles using RL to explore various tracks. After testing *ARtonomous* with middle school students, they found that it ignited student curiosity about RL/AI.

It is observed that the students were engaged through these projects and tools, and they increased curiosity among middle school children to further explore RL. Both *PlayGrid* and *ARtonomous* introduce children to RL using visualizations. We adopted a more interactive approach, similar to *ArtBot*. TrainYourSnakeAI introduces children to reward functions, a critical component of RL algorithms, through the simple and familiar Snake game. We decided to use a game most children are familiar with to add excitement and keep them engaged. Additionally, TrainYourSnakeAI collects quantitative data while users play the game, addressing a gap in many previous studies which primarily use qualitative data to measure student learning outcomes.

3 Design of TrainYourSnakeAI Game

Our web application exposes children to RL. We chose RL because we thought it would be an intuitive concept for students to understand, while also showing them an important concept in AI. While designing our application, we kept in mind two of the "Five Big Ideas in AI" [22]: Representation and Reasoning, and Learning. The main idea of our software is for children to train their own AI to play the game Snake and observe how the snake reacts based off different reward functions specified by the user.

3.1 Game-play

Our application first asks the user to input their first name, last initial, and grade level to capture data. We then give them a brief introduction to RL. Then, the children are asked to briefly play the Snake game themselves. Finally, they are taken to a page called the "Training Arena" where they can play around with different reward functions for the snake and train it in different ways.

3.1.1 Introduction to RL. Once the child enters their information, they are taken to a page where they are briefly introduced to RL. This page has text explaining that "If you reward a chicken when it does a back flip, it will do more back flips. Also, if you punish a chicken when it misbehaves, it won't misbehave as often. Think of training your AI like this. This method is called RL. You will be training your AI to play the game Snake. But first…Play the snake game yourself!." Additionally, we added an image of a chick to this page to grab the attention of the children and make the software more playful and relaxed. The introduction paragraph is not a formal definition of RL by any means but compares RL to how a chicken (or any biological organism) learns. The idea was to provide an easy-to-understand and intuitive definition for children, while not straying from the formal definition.

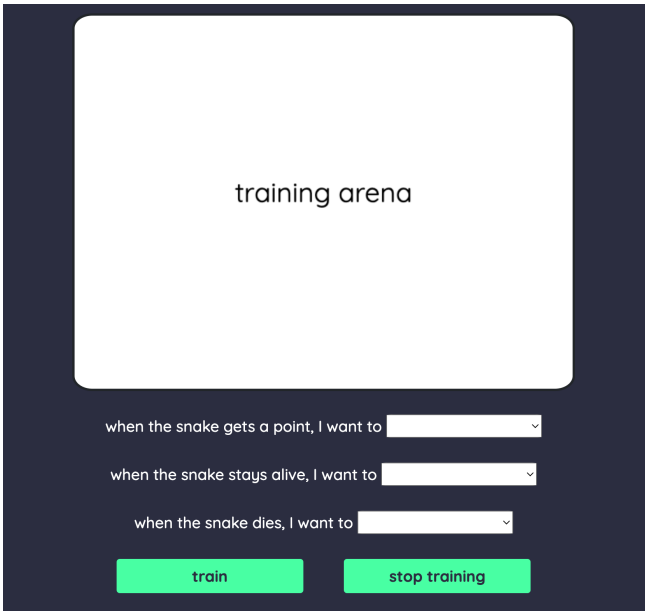

Figure 1: Training Arena

3.1.2 Snake. The Snake game with RL involves an agent that learns to play by interacting with the game environment, receiving rewards and penalties, and continuously improving its strategy through repeated play and learning algorithms [21, 26].

The game environment includes the snake, food, and the game board. The snake's objective is to eat the food that appears randomly while avoiding collisions with itself and the board's boundaries. An RL agent learns to play the game by interacting with the environment, receiving the current state, and taking actions (moving up, down, left, or right). The agent gets rewards for eating food and penalties for collisions. It develops a policy to choose actions based on the state and uses algorithms such as Q-learning to learn and improve its policy over time. By playing numerous games and updating its policy, the agent learns to make better decisions, such as finding efficient paths to the food and avoiding collisions.

Nokia is well known for including the Snake game on its phones starting in 1997 [4]. Though the game's concept dates back to arcade games in the 1970s, the Nokia version played a key role in popularizing it globally [6, 16]. The highly successful Nokia 3310, launched in late 2000, further cemented Snake's iconic status. The game became a cultural phenomenon and has inspired many remakes and adaptations for modern devices[20].

We decided to let the children play the Snake game themselves for those who are not familiar with it or who have not played it recently.

3.1.3 Training the AI. The "Training Arena" (shown in Figure 1) is where the core of our application lies.

The user sees a familiar box, which is identical to the rectangle screen in the previous page where the user played Snake. However, this time they are prompted with three fill-in-the-blank questions. The questions read as follows:

- when the snake gets a point, I want to *blank*
- when the snake stays alive, I want to *blank*
- when the snake dies, I want to *blank*

There is a drop down box for each question where the user chooses one of three options to fill in the blank; the options are as follows: "reward it", "leave it alone" and "punish it." An example response is as follows:

- when the snake gets a point, I want to *reward it*

We chose this fill-in-the-blank format to provide the user with as much context as possible.

Once the user makes their selections and clicks "Train," the fill-in-the-blank values "reward it", "leave it alone" and "punish it" are translated to a +10, 0, or -10 respectively. They are then fed into a brand-new RL agent. This agent will train for a total of 100 games. The fill-in-the-blank values are given to the agent's game state vector, which is used by the AI agent to represent its environment. The training happens in real-time, and the user is able to see the training on the screen. In this way, the children can see the effect of their reward functions on the agent's behavior. If the children train the AI correctly, it would be as follows:

- when the snake gets a point, I want to *reward it*
- when the snake stays alive, I want to *leave it alone*
- when the snake dies, I want to *punish it*

The snake would begin to do well around game 75. After this, it becomes clear that the user has correctly trained the agent as the snake can reach scores above 40 as shown in Figure 2.

On the other hand, the user can provide a negative configuration as show below:

- when the snake gets a point, I want to *reward it*
- when the snake stays alive, I want to *reward it*
- when the snake dies, I want to *punish it*

The agent then begins to learn that by staying alive, it receives a reward and chooses to go in circles instead of going for the food. The user is able to witness this behavior that comes as a direct result of how they trained it.

If the user decides to go for a very negative configuration, it could be as follows:

- when the snake gets a point, I want to *punish it*
- when the snake stays alive, I want to *leave it alone*

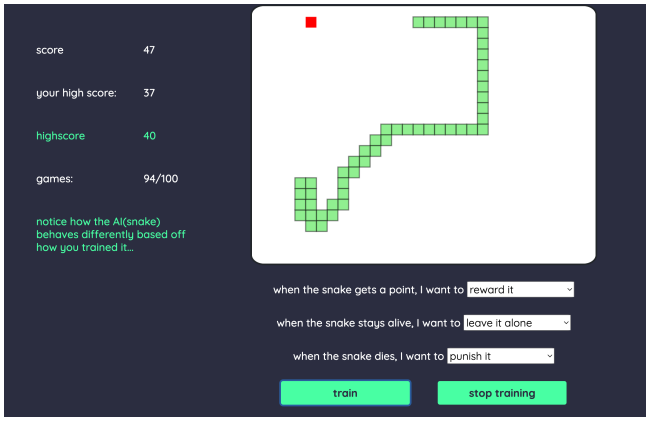

Figure 2: Snake AI with a high score

- when the snake dies, I want to *reward it*

The user can see that the snake begins to learn that by dying, it receives a reward. Therefore, after a few games, the snake simply goes straight for the wall and ends the game intentionally.

Our application helps children understand that after exploring its environment, the agent will perform actions based on what the user defined as the reward functions. The children will see that the agent increasingly tries to get rewarded and avoid punishment as it learns.

3.2 Technical Details

Our application is built with Python using Flask as our web framework with a SQLite database.

The RL agent that we used is from [17] with some modifications. By dynamically changing the agent's reward functions (i.e., a variable for reward, a variable for staying alive, and a variable for dying), the AI exhibits a drastic change in the learned behavior, and we used this as the premise of our application.

To provide an overview on the technical aspects of the AI agent being used in our software: A state vector representing the current snake's environment is given to the model's predict function, where we are given back an action to take, also a vector. This action is then performed inside the game and the result is evaluated. There is now a different state vector for the environment, since an action was just performed inside the game.

The model then updates based off of these variables so that the next time it is called to predict, it now has more information on the best move to take. The model uses a Feed First Neural Network with 11 input variables (a state vector representing the current environment), 256 hidden layers, and 3 output layers (the action vector giving us the optimal action to take). This model is trained repeatedly, with the goal to train it sufficiently to where you provide a current environment (state vector with 11 values) and it will give you an optimal action to take (action vector with three values) based off its environment.

When the user initializes a training, the AI agent begins training on the server side. We facilitate the real time communication from the server to the client with web sockets. Streams of data are sent from our backend to a JavaScript frontend, allowing the user to see the training process in real time.

The source code for our application can be found at [9].

4 Study Design

This web-based application *TrainYourSnakeAI* was tested at a public STEM charter school in a major city in the state of Texas. The participants were middle school students (in 6th through 8th grade) who signed up and attended the after-school sessions. There was a total of 122 participants. However, of those 122 children, only 71 of them used our software due to time constraints. This study consisted of two 3-hour sessions over the course of two days. This research took place in what we called an "AI expo," where there were seven applications (including *TrainYourSnakeAI*) that exposed children to different concepts in AI. There were usually 3-7 students per table where an application was being demoed. Once the children were done using a project, they would rotate on to the next one. Each student used the software for about 5-10 minutes. We have IRB approval for this work and we obtained parental consent and student assent.

Table 1: Grade level distribution of the participants

Grade Level	Number of Students	Percentage
6	23	32.4
7	30	42.3
8	18	25.4

4.1 Learning Activities

In our table for this software, we had four computers running our application locally. The authors spent time with the children discussing the software and provided guidance if needed. The children were first briefly introduced to the application by one of the authors. Afterwards, they started using the application. Once they were done, the children completed the post-survey form.

4.2 Data Gathered

We collected data from the following sources: pre-survey, post-survey, screen/audio recording, and automated data collection from our software. The pre-survey consisted of a universal questionnaire that was filled out by all the students who participated in the "AI expo" study. The pre-survey consisted of questions that gathered data on their current understanding of AI. The post-survey we used was designed specifically for our application and was given to only the students who interacted with our software. Screen/audio recordings recorded participant interaction with our application; however, these were not used in the analysis of our research. Finally, we also gathered application-specific data by saving the following information into our database: user information and metrics on the AI models (snakes) that the user created. The application-specific data provided statistical information on the use of our application, not student learning.

5 Results

We focused primarily on the results of the post-survey because we believe that they brought the most insight from the different sources of data we gathered. Moreover, we also connected the pre-survey data to the post-survey data to evaluate the conceptual difference in understanding RL.

5.1 (RQ1) How can we create an engaging tool to teach reinforcement learning?

Our research study was aimed at introducing AI concepts to children and we believe it was also important to try and ignite curiosity and excitement along the way. The results of the first item of the post-survey questionnaire (shown in Figure 2) were used to measure student satisfaction with our application. 81.7% of the students responded "Yes," that they had fun training their own AI.

Table 2: Responses to the post-survey item "Did you have fun training your own AI?" The options were "Yes," "Meh," and "No."

Selected Response	Percentage	Count of Students
Yes	81.7%	58
Meh	18.3%	13
No	0%	0

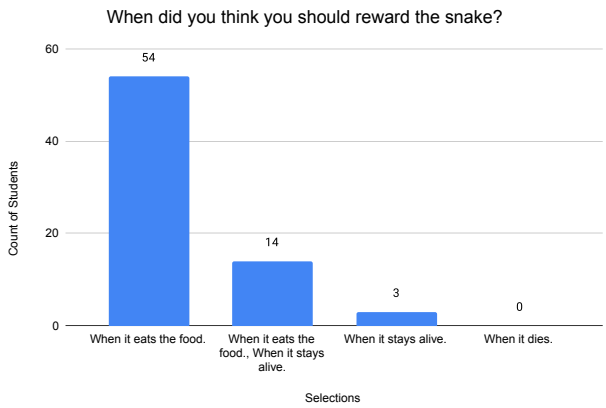

Figure 3: Responses to "When did you think you should reward the snake?"

5.2 (RQ2) Does using our application foster a stronger understanding of reinforcement learning in children?

To measure the student's understanding of RL, we first analyzed the responses to "*When did you think you should reward the snake? (choose any number of options).*" The correct answer to this item is "when it eats the food." Students were allowed to choose multiple responses and 54 correctly selected only the correct answer, as shown in Figure 3. An additional 14 students selected "when eating the food," but also incorrectly selected "when it stays alive."

Also, another item asked, "*Do you think RL is similar to how a dog learns?*" We found that 64 of the 71 students reported that RL is similar to how a dog learns. We are aware that this question could have been positively biased given the introduction of RL that was provided to them in the "Introduction" page of our application. However, we believe that it was a combination of all the aspects of our software that instantiated these responses.

Next, we directly mapped our final post-survey item "*I would describe RL as...*" to pre-survey responses as we felt the combination of the open-ended nature of this question and the data from the pre-survey would provide insight. The open-ended responses were coded into four categories as follows:

- Reward and/or punishment: This is the most accurate of the four categories. The children answered the question correctly by describing RL with the keywords reward and/or punishment.

Table 3: "No RL" group describing RL

Codes	Number of responses
Total	51
Reward and/or punishment	21
Natural learning	19
Fun	4
Off-Topic/No Response	7

Table 4: "RL" group describing RL

Codes	Number of responses
Total	20
Reward and/or punishment	8
Natural learning	10
Fun	0
Off-Topic/No Response	2

- Natural learning: The response came in the form of analogy, more specifically comparing RL to the learning of biological organisms. We considered these responses accurate.
- Fun: Rather than answering the question as intended, the children expressed their feelings toward performing RL on an AI agent.
- Off-topic/No Response: Responses that were off-topic, inaccurate, or empty.

The following are some examples of responses from children that fell into each of these four categories:

- Reward and/or punishment: "RL is teaching a machine to know what to do more (which is good) and what to do less (which is not good) by giving them rewards when they achieve something."
- Natural learning: "giving a dog a treat when he behaves and saying bad when he misbehaves"
- Fun: "interesting and cool"
- Off-Topic/No Response: "Idk"

We separated the children's data into two groups: those previously familiar with RL and those who were not already familiar with RL. We separated the children's data based on their answers to the following pre-survey multiple-choice question: "*I am familiar with these specific AI concepts (check any that apply).*" If the child selected "RL" as an answer, they were grouped into the "RL" category, and those who did not mark "RL" were added to the "No RL" category. We found that 51 out of the 71 students were not previously familiar with RL. We analyzed the results of our open-ended question with the "No RL" group to identify if they could accurately describe the AI concept after using our application.

Of the 51 students previously unfamiliar with RL, the 21 responses fell into the category "reward and/or punishment" and 19 fell into "natural learning." In this group, four responses fell into the category "fun," and seven responses fell into "off-topic/no response."

We performed a Fisher's Exact test on the results in Tables 3 and 4 to see whether there is an association between previous knowledge of RL and performance on this open-ended post-survey item.

For each group, the responses in the categories "reward and/or punishment" and "natural learning" were classified as correct and responses in "fun" and "off-topic/no response" were classified as incorrect. As such, there were 40 correct responses and 11 incorrect responses in the "No RL" group, and there were 18 correct and 2 incorrect responses in the "RL" group. The Fisher's Exact test yielded a p-value of 0.326, indicating there is not a significant association between previous knowledge of RL and the correctness of the response to the open-ended item.

6 Discussion

Overall, most students had fun interacting with *TrainYourSnakeAI* as indicated by the responses to the post-survey item "*Did you have fun training your own AI?*" Most of the students (81.7%) responded that they had fun. Furthermore, none of the students responded "no." This suggests that most of the students enjoyed interacting with our application.

As seen in Table 3, from the "No RL" group, 21 students were able to accurately describe RL using the keywords rewards and/or punishment. We also found that 19 children compared RL to the natural learning process of biological organisms. We include those 19 children as having described RL accurately. The purpose of our application was to expose to children RL and we found that 40 of the students who were not previously familiar with the AI concept were able to describe it to a certain extent. If we move our attention to the category "RL" (students who were already familiar with RL before interacting with our application), we see that there were a total of 18 students who accurately described RL out of 20.

We performed a Fisher's Exact test for the results because the data did not fulfill the conditions of the two-proportion z-test. The results indicated that there were no significant differences between the "RL" and "No RL" groups. Therefore, it seems that both groups gave similar amounts of responses in regard to accurately describing RL. We believe that there can be no concrete conclusions with comparing the tables of data. The pre-survey question used to categorize if the child was familiar with RL was "yes or no" fashion. More insight would have been achieved if the question had been in the open-ended manner of the post-survey question used to describe RL as shown in Tables 3 and 4 to truly gauge their knowledge of the AI concept. However, if we focus on children unfamiliar with RL beforehand, there was a significant amount of them who gave an accurate definition of RL post-application use. Although the concept of RL and reward functions might still be abstract to these individuals, we believe that this application provided them with an intuition in this AI concept.

7 Ongoing and Future Work

During the conduct of this *TrainYourSnakeAI* application with middle school students, we recognized that the Snake RL agent did not show visible improvement until about iteration 75 of 100 of the training epoch. To mitigate this limitation, we added a live visualization of the state of the game as viewed by the RL agent, so that participants can see what the AI snake is "thinking" as it is being trained. A "This is what the snake AI is thinking" panel was added to the user interface of the "Training Arena" (Figure 4). This version of our application needs to be tested with more students before we have meaningful results.

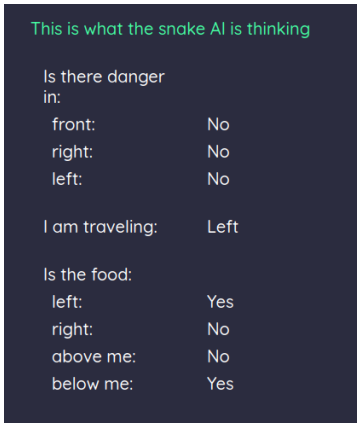

Figure 4: "What the Snake AI is Thinking" panel added to Training Arena

This is an ongoing work. This updated version of our application needs to be tested with additional students to determine its impact on their learning. We also wish to increase the sample size of in a future study to better understanding student learning impacts.

Furthermore, we can also integrate other RL concepts and make the application available through web browsers to make it even more informative for children and easy to use for K-12 teachers.

8 Conclusion

In this work, we introduced Reinforcement Learning to middle school children by giving them the opportunity to train their own AI model. Although we cannot conclude that there was any concrete difference in understanding AI concepts, we can observe that 40 out of the 51 children not previously familiar with RL were able to accurately describe it after using our software. Additionally, most of the students indicated that they enjoyed training their own AI using this *TrainYourSnakeAI* application. This work is part of an effort to expose children to AI education and AI literacy.

Overall, while RL is a complex subject, efforts to introduce it at various educational levels are increasing, supported by initiatives that provide age-appropriate, engaging, and accessible learning experiences. These initiatives highlight the importance of early exposure to AI concepts to build a foundation for future learning and innovation in the field.

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